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HUONG LE, Hanoi University of Science and Technology, Hanoi, Vietnam

NGOC LUU, Hanoi University of Science and Technology, Hanoi, Vietnam

THANH NGUYEN, Hanoi University of Science and Technology, Hanoi, Vietnam

TUAN DAO, Hanoi University of Science and Technology, Hanoi, Vietnam

SANG DINH, Hanoi University of Science and Technology, Hanoi, Vietnam

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HUONG LE, Hanoi University of Science and Technology, Hanoi, Viet Nam

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The development of large language models (LLMs) such as ChatGPT and Gemini has led to impressive advancements in question answering (QA) systems. However, they often rely on generic knowledge from the internet, resulting in hallucinated answers when applied to domain-specific QA tasks. Furthermore, their operational dependence on powerful GPUs poses challenges for practical software deployment. Building a QA systems for low-resource languages like Vietnamese is even more challenging due to the scarcity of labeled data and limited pre-trained language models. In this study, we aim at constructing a Vietnamese legal QA system using a retrieval-augmented generation approach to reduce incorrect outputs. Our focus is on improving answer generation accuracy by training small-scale LLMs suitable for real-world deployment. Our contributions are: (i) constructing Vietnamese legal provisions and QA datasets for training the system; and (ii) proposing methods to fine-tune language models with QA capabilities in the legal domain. Experimental results demonstrate that it is possible to train an LLM with fewer computational resources and a smaller dataset while maintaining effectiveness. Our findings highlight that designing an efficient training and fine-tuning strategy is crucial for overcoming these challenges, particularly in the context of Vietnamese legal question-answering tasks.

CCS Concepts: • **Computing methodologies** → **Natural language generation**;

Additional Key Words and Phrases: Vietnamese, Legal, Question Answering, answer generator, BartPho, VinaLlama

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Authors' Contact Information: Huong Le, Hanoi University of Science and Technology, Hanoi, Viet Nam; e-mail: huonglt@soict.hust.edu.vn; Ngoc Luu, Hanoi University of Science and Technology, Hanoi, Viet Nam; e-mail: ngoc.lt204595@sis.hust.edu.vn; Thanh Nguyen, Hanoi University of Science and Technology, Hanoi, Viet Nam; e-mail: thanh.nx204692@sis.hust.edu.vn; Tuan Dao, Hanoi University of Science and Technology, Hanoi, Viet Nam; e-mail: tuan.dm231106m@sis.hust.edu.vn; Sang Dinh, Hanoi University of Science and Technology, Hanoi, Viet Nam; e-mail: sangdv@soict.hust.edu.vn.

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1 Introduction

Question answering is the task of providing answers to natural language questions based on available knowledge or data. Traditional QA systems rely on techniques such as question analysis, information retrieval, and text extraction to identify the most relevant answer from a large text corpus [3, 33, 40]. However, these approaches have some limitations such as their ability to handle only specific question formats, relying on syntactic parsers, pre-defined question patterns, and restricting answers to either a single value or a snippet in the input text passages. This approach lacks the capability to fully understand natural language and process diverse questions.

The performance of question-answering has greatly improved due to rapid development and recent breakthroughs in deep learning. Deep learning models are capable of identifying the semantic relationships in questions and generating meaningful answers based on patterns learned from vast amounts of data. ChatGPT [11], Gemini [6], and Claude [40] represent notable examples of these models. Their large size and extensive training on a vast amount of text and tasks enable them to perform well across multiple languages. However, these models are prone to AI hallucinations due to the pre-trained datasets being gathered from various sources, not focusing on any specific domain. Additionally, their reliance on powerful GPUs makes them expensive to operate, limiting their accessibility for small businesses with constrained budgets. Meanwhile, there is a significant demand from businesses and users for domain-specific question-answering systems, such as those focused on areas like government law, banking, insurance, and education. These systems need to provide accurate answers based on information within their respective fields. Such an application in Vietnamese-specific domains remains limited due to the scarcity of Vietnamese training data.

To address the AI hallucination problem and improve model accuracy and reliability, several QA approaches rely on **retrieval-augmented generation (RAG)** techniques, with the answer generator fine-tuned on a well-pretrained language model. With RAG, the AI model generates answers using the information retrieved by the retriever module rather than relying on the model's pre-training data.

In text retrieval, the main challenge is to ensure that the correct text passages are ranked within the top-k search results, where k is typically 1, 5, 10, or 100. Meanwhile, the challenge in answer generation tasks is to be able to interpret the input text and produce accurate, concise, and informative answers without making up information. Most recent approaches to answer generator using small language models with hundreds of millions of parameters involve extracting a snippet from the input passages. However, in the realm of legal texts, users often seek precise answers to their inquiries rather than extracting a snippet from legal documents. This task poses a significant challenge as it demands a comprehension of the input text and the ability to employ reasoning mechanisms to generate appropriate responses. While the text retrieval task has been researched for quite a long time, there is only a few research on the task of answer generation using small-scale LLMs, presenting numerous challenges for researchers. In this context, small-scale LLMs refer to language models with parameter counts ranging from several million to several billion. In this article, we propose a question-answering model that focuses on enhancing the accuracy of answer generation for Vietnamese legal question answering.

To address this task, we leverage Vietnamese LLM models proficient in processing and generating Vietnamese text. Guided by the criteria of practical applicability, meaning the system does not require powerful GPUs, runs efficiently, and produces reasonable results, we propose utilizing two models: BartPho-Large [35], and VinaLlama [23]. These models will be trained on various datasets for comparative analysis and evaluation.

The key contributions of our work are summarized as follows:

- **Construction of two datasets:** We have constructed two legal text datasets: vnLawData and vnLawQPA legal question-answering dataset. The vnLawData dataset comprises an extensive collection of 21,000 legal documents spanning 25 diverse domains. This rich and varied dataset accurately reflects the breadth and complexity of the Vietnamese legal system. The vnLawQPA dataset, on the other hand, encompasses a substantial collection of 200,000 question-passage-answer triples specifically designed to facilitate the fine-tuning of the legal question-answering system.
- **Fine-tuning Scenarios:** We propose ten scenarios to fine-tune small-scale LLMs for the answer-generation task, aiming to enhance system performance in domains with limited training data while minimizing computational resources. Our study evaluates the impact of utilizing related-task data for training question-answering models by analyzing the effectiveness of various data augmentation scenarios.

2 Related Works

2.1 Datasets

There have been several QA datasets proposed in recent years; some contain answers as snippets from the input text passages, while others have answers generated based on reading comprehension. Examples of datasets with extractive answers include SQuAD [30], RACE [15], and Kien et al. [13]. The SQuAD dataset [30] contains over 100,000 questions derived from Wikipedia articles, with each sample consisting of a question, a context paragraph, and an answer that is a specific span from the context. RACE [15] is a large-scale multiple-choice from English examinations in China. RACE is notable for its significantly longer contexts question dataset, comprising over 28,000 passages and nearly 100,000 questions sourced compared with other popular reading comprehension datasets. Also, RACE has a high proportion of questions requiring reasoning. Kien et al. [13] developed a question-answering dataset based on Vietnamese legal documents, containing 5,922 queries and their corresponding articles as answers; however, this dataset is not publicly available.

Some datasets featuring text generation answers include DISC-LawLLM [41], Bactrian-X [20], and P3 [31]. DISC-LawLLM dataset [41] was constructed by crawling real-world legal text and consultation data from judicial advisory websites. It includes 93,000 legal QA pairs (questions and answers) and 23,000 legal QA triples (question, passage, and answer), all in an instructional format. Similarly, the Bactrian-X dataset by Li et al. [20] is instruction-based, comprising 52,000 samples and 15,000 samples translated into 51 different languages from subsets of Alpaca and Dolly general tasks, respectively. P3 (Public Pool of Prompts) dataset [31] is a collection of prompted English datasets covering a diverse set of NLP tasks, including QA. This dataset contains approximately 350,000 samples, divided into four subsets corresponding to four tasks: text summarization, answer extraction, answer selection, and input-based context generation.

To the best of our knowledge, there is currently no available dataset for Vietnamese question-answering tasks where answers are generated based on the related context. In our work, we have constructed such a dataset.

2.2 Question-Answering Models

Traditional approaches for question answering rely on natural language processing techniques such as predefined rules and patterns to get the appropriate documents related to the query and to generate the answers [14, 32, 39]. For example, Kim et al. [14] implemented a system to answer

yes/no questions on legal bar exams using two stages: (i) identifying relevant legal documents and (ii) analyzing them to find answers. To address the second task, they conducted linguistic analysis to match predicates and arguments between the questions and the knowledge source sentences. These methods lack the ability to adapt to new patterns and changes in the data.

With the advent of deep learning, QA has made considerable progress. Liu et al. [21] introduced RoBERTa, an improved version of the BERT model [5], which can process text without relying on predefined patterns. The RoBERTa model achieved state-of-the-art results on the MRC task with SquAD and RACE dataset.

LLM systems such as chatGPT [11], Gemini [6], and Llama3 [18] can answer a wide range of questions, no longer relying on pre-defined patterns and can generate answers based on text comprehension rather than extracting from the original text. These systems are based on LLMs such as GPT-3 [1] that utilize advanced learning algorithms and are trained on massive amounts of data with the goal of enabling the LLM to interpret language and generate accurate answers. However, these models require substantial computational resources for training and operating, massive amounts of data for training, and often generate hallucinations.

These systems are based on LLMs such as GPT-3 [1] that utilize advanced learning algorithms and are trained on massive amounts of data with the goal of enabling the LLM to process language and generate accurate answers.

Domain-oriented question-answering systems based on small-scale LLMs often focus solely on the information retrieval module [3, 8, 12, 29, 37]. This approach does not provide specific answers, and users must read the top-k results to extract the information they seek. Duong and Ho [3] constructed a legal question-answering system based on searching legal documents using query phrases consisting of verb phrases and noun phrases representing the text. To improve search accuracy, the system expands the search by including synonyms for these identified phrases. The next step is to find relevant text passages in the top 10 returned by measuring the similarity between the question and these articles.

Other studies train pretrained LLMs to generate more concise answers based on the provided context but are limited to certain simple formats such as extractive, yes/no, or multiple-choice [4, 27]. Extractive answers are often lengthy and sometimes do not directly answer the question. Dao and Le [4] built a Vietnamese multiple-choice question-answering system in the biology domain using an LLM. Nguyen et al. [27] built a question-answering system that can answer three types of questions: Yes/No, multiple-choice, and factoid questions. For factoid questions, they extract a passage from the text to answer the question. The question-answering dataset was collected from legal websites but is not publicly available.

3 Methodology

In this work, we aim at developing a Vietnamese legal question-answering system that generates concise answers that directly address the query. We employ small-scale LLMs with a maximum of 7 billion parameters to ensure practical applicability. To prevent hallucination, we adopt the RAG approach from LangChain [16] and customize it to suit our task. The architecture of our Vietnamese legal QA system, depicted in Figure 1, consists of two primary modules: (i) the information retriever and (ii) the answer generator.

In our system, the original legal documents were segmented into passages for two reasons:

- **Efficient Passage Identification:** Returning entire documents as search results would require an extra step to locate the specific passages relevant to the query within each document. By dividing the documents into passages, the retriever can directly return the relevant sections, simplifying the retrieval process. From observing questions and answers

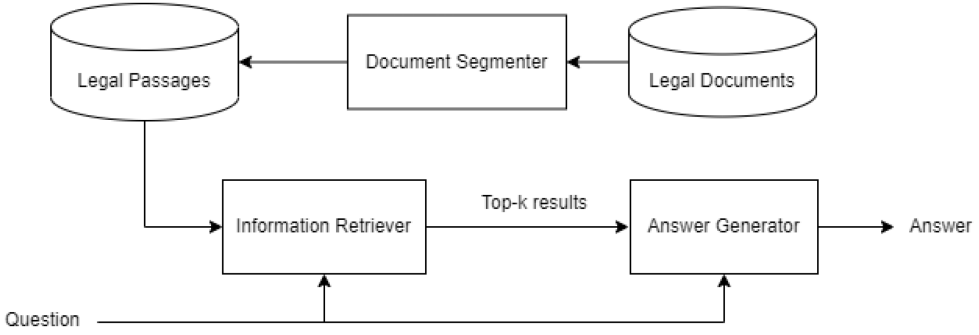


Fig. 1. Architecture of the Vietnamese Legal Question-Answering System.

in the legal domain, we found that most answers are contained within passages no longer than 256 tokens. Therefore, we split each document into passages of up to 256 tokens.

- LLM Input Limitation:** Answer generators that rely on language models for text comprehension are constrained by input size limitations. The answer generator takes both the question and the context as input. The context is formed by combining multiple retrieved passages until the input size limit is reached. Segmenting documents into passages of up to 256 tokens allows multiple passages to be combined within the input limit of LLMs.

The information retriever searches for the most relevant passages from a Vietnamese legal text data repository. To achieve this, we employ a combination of bi-encoder and cross-encoder techniques over the legal text database. The bi-encoder and cross-encoder techniques involve encoding both the query and the passages into vector representations and then calculating the similarity between the vectors. The similarity score indicates the degree of relevance between the query and the passage.

The top-k results from the information retriever are used as input for the answer generator. Since the F-score of the top-10 results in our system exceeds 90%, the value of k is set to 10. This ensures that the passages in the top-10 are the most relevant ones, serving as context for the next step in answer generation.

The answer generator relies on the top-k retrieved passages to produce concise and accurate responses that directly address the user's query. The precision of the answer generator directly influences the overall accuracy of the QA system. If the answer generator successfully identifies the correct answer within the top-k retrieved passages, the overall accuracy of the system will be higher than the top-1 accuracy of the information retriever. This highlights the importance of the answer generator in extracting the most relevant information and formulating an accurate response.

Due to the input size limitation of LLMs, the top-k passages are divided into several sets of segments, each serving as a context for the answer generator. The generated answers from those sets are then combined to be used as context for the final run of the answer generator, producing the final answer.

Segmenting the legal documents into passages might cause answers to be scattered across multiple segments. To resolve this, we identify cases where multiple retrieved passages within the top-k results are adjacent in the original document and merge them into a single set before processing by the answer generator. This approach ensures that the answer generation process takes the full context into account, producing a complete response.

Based on the RAG architecture of LangChain [16], this article focuses on proposing methods for fine-tuning LLM models that use fewer computational resources to build an answer generation

model for domains with limited training data. Additionally, this article provides a more detailed discussion on document-splitting methods for retrieval and propose effective strategies for handling adjacent passages within a document that appear simultaneously in the top-k search results.

The answer generation task can be formalized as follows:

Given a query Q and m relevant text passages $P = p_1, p_2, \dots, p_m$ (where m represents the m passages within the top-k retrieval results), the task is to generate an answer A based on the question and the information contained within these m passages.

The fundamental challenge for the answer generator lies in selecting pretrained models capable of interpreting Vietnamese, the quality of training datasets, and a reasonable training process. In the following section, we will propose our solution to solve the above-mentioned problems.

3.1 Model Selection

Given the practical applicability constraints of our system, which include running efficiently on a low-configuration GPU (an RTX 3090 with 24GB VRAM is used in our experiments) and producing reasonable results, we implement the fine-tuning of two seq-2-seq LLMs, BartPho, and VinaLlama-7B, for comparison and evaluation.

BARTpho, developed by Tran et al. [35], is a sequence-to-sequence model based on the Bart "large" architecture [17] with 420M parameters and the maximum input length of 1024 tokens. BARTpho was pre-trained on 150,704 news articles in a Vietnamese text summarization dataset [22] and then fine-tuned on the same dataset for the text summarization task. In both automatic and human evaluations, BARTpho outperforms mBART (Multilingual BART) [26] in text generation tasks and achieves state-of-the-art performance for Vietnamese text generation.

The BartPho model is compact and easy to deploy. However, small models have some limitations compared with large models, such as less ability to reason and paraphrase answers. They often extract answers from the context instead of rewriting them, while humans typically prefer simple and easy-to-understand answers. Therefore, we carried out other experiments with VinaLlama-7B to evaluate how a larger model impacts the system's output.

The pretrained VinaLlama-7B [23], a large language model designed for the Vietnamese language, is built upon the pretrained Llama-2-7B [36] and further trained on an additional 800 billion tokens from Vietnamese books and newspapers. VinaLlama-7B-chat [23], a full fine-tuned version of VinaLlama-7B trained on 1 million samples from various tasks such as text generation, machine translation, question answering, and text summarization, is capable of handling QA tasks. However, since VinaLlama-7B-chat has not been trained on legal data, we selected the pretrained VinaLlama-7B for further training on legal domain data to develop a model specifically suited for legal question-answering tasks.

3.2 Datasets

Dataset for continue pre-training:

The Vietnamese legal provisions are taken from the official legal library website named Thuvien-phapluat [34], containing 21,000 documents. These documents cover 25 different fields, reflecting the diversity and richness of the Vietnamese legal system. This corpus is called **vnLawData**. It is used to continue pre-training the two LLMs used in our experiments.

Law database: The knowledge for the information retrieval is taken from vnLawData. To construct the searching database for the information retrieval, these legal documents are first segmented into articles with a maximum length of 256 tokens for ease of content delineation for answering. There are cases where an article's length is more than 256 tokens, in that case, the article will be split into two or more passages. All articles and passages are attached with their corresponding number and title of the legal document. During the inference process of the retriever,

if the top-k retrieved passages contain several continuous passages of the same legal document, these passages will be concatenated and included as the input context of the answer generator in order to ensure the complete answer is preserved.

Dataset for finetuning:

To effectively fine-tune the pretrained LLMs for Vietnamese legal question answering, a legal QA dataset is needed, where each sample consists of a question, context, and answer. The *Thuvienphapluat* website [34] contains a legal Q&A section with questions from Vietnamese citizens and answers provided by lawyers. Therefore, we extract tuples of (question, the passage containing the answer, and the corresponding answer) from this website. For each question, we construct a context comprising three passages:

- **Answer Passage:** The passage containing the answer for the input question.
- **Related Passages:** Two additional passages incorporate keywords from the input question, but these passages do not contain the answer. If the Answer Passage is long, we will reduce the number of Related Passages added to the context. This mimics the input format encountered during the inference phase of the answer generator.

The final legal QA dataset contains 150,000 samples, and is named **vnLawQPA-pos**. A data sample of the vnLawQPA-pos dataset is provided in Appendix A.1. A disadvantage of this dataset is that it contains a higher proportion of answers that involve direct extraction from the context compared with those requiring reasoning. To address this problem and to enhance the system's ability to interpret and infer from legal texts, we plan to incorporate additional datasets alongside vnLawQPA-pos.

To effectively train the system to handle cases where the input context of the answer generator does not contain the answer, we generate a set of 50,000 negative samples alongside the positive samples. The context passages in the negative samples contain keywords from the question but do not include the answer. In this case, the output answer should be "Sorry, I don't have information to answer your question", in order to train the system not to produce incorreced answers. These samples are created by replacing the context by randomly taking three other passages from the same legal document. This set is called **vnLawQPA-neg**.

The combined dataset of vnLawQPA-pos and vnLawQPA-neg is called **vnLawQA** with 200,000 samples.

Evaluate set: We create 2000 samples for the test dataset using the same method as creating positive and negative samples as above, in which the test data does not overlap with the training data. The test dataset includes several types of samples: those with context containing answers, those with context lacking answers, those with context containing extracted answers, and those with context containing answers that require inference.

As mentioned earlier, the vnLawQA-pos dataset does not have enough samples for training reasoning ability, we need other datasets for this purpose. However, there is not much Vietnamese dataset for this purpose. To enhance data, in addition to collecting QA samples from available websites, we translated related datasets from English to Vietnamese using the Google API tool. The augmented datasets used in training the LLMs are as follows:

- **vnGeneralService:** Gather 10,000 question-answer pairs from the *Dichvucong* [2] website. These samples lack contextual passages, but the answer generator can utilize the 21,000 legal documents from vnLawData to generate responses. The goal is to train the LLMs to effectively interpret questions and retrieve answers from the model that is pre-trained with legal documents.

- **DISC-LawLLM dataset** [41]: Extract 40,000 samples from the DISC-LawLLM dataset, focusing on tasks related to content summarization and context-based answering.
- **Bactrian-X dataset** [20]: Utilize 72,000 samples from the Bactrian-X dataset, specifically from the Alpaca (52,000 samples) and Dolly (15,000 samples) subsets, which serve for general tasks.
- **P3 dataset** [31] containing approximately 350,000 samples. This dataset contains four subsets corresponding to 4 tasks: text summarization, answer extraction, answer selection, and input-based context generation.
 - Text summarization: This task involves generating a concise summary of a longer piece of text. The answer extraction task
 - Answer extraction: This task involves identifying the passage of text that contains the answer to a given question.
 - Answer selection: This task involves selecting the correct answer from a set of multiple-choice options.
 - Input-based context generation: This task involves generating additional context for a given piece of text.
- **Vi-Alpaca dataset** [25] containing 75,000 samples: The generation of this dataset is mimic the Alpaca dataset by taking data from Vietnamese news sources and using instruction samples in Ref. [35] to create data for many different tasks. Data is generated using ChatGPT.

In the following sections, when referring to fine-tuning the LLM using datasets, it is understood to use the translated version of those datasets as mentioned above.

According to [7,10], continual learning of new natural language tasks poses the challenge of catastrophic forgetting, where the system gradually loses knowledge of previously learned tasks. Harang and Sanders [7] also pointed out that models trained on all task-specific datasets simultaneously consistently outperform those fine-tuned sequentially. Therefore, to prevent the catastrophic forgetting problem, all subsets of the dataset are combined during the fine-tuning process in our experiments to retain knowledge from earlier tasks.

3.3 Training LLM Models

Bartpho is trained on Vietnamese news data, while Llama is trained on a multi-domain, multi-language, including Vietnamese. Neither of these models has been trained extensively on legal domain data. To make the LLMs adapt with legal domain, we conduct continual pre-training the two models with the vnLawData dataset. For the Bartpho model, we updated the weights on the entire model. For the VinaLlama-7B model, due to hardware limitations, we used the QLoRA method (batch size=4, rank_lora=16) to reduce the cost and time of model training.

After continual pre-training, we fine-tuned the BartPho and VinaLlama models using labeled data across various scenarios. Given that the dataset of 150,000 question-answer samples is relatively small for fine-tuning a large language model, we incorporated additional training data from available sources (as outlined in Section 3.2) to enhance the system's training. We experimented with several scenarios to evaluate the impact of fine-tuning the models on diverse task-specific datasets and to determine the optimal solution for the Vietnamese legal QA task. The fine-tuning process was carried out under the following scenarios:

Scenario 1: Fine-tune the Bartpho model using 150,000 samples of Vietnamese legal question-answering data (vnLawQPA-pos).

This scenario is used to evaluate the BartPho model using only the legal question-answering dataset **vnLawQPA-pos** without incorporating any external augmented data.

Scenario 2: Fine-tune the Bartpho model using 350,000 samples from the P3 dataset, followed by fine-tuning with 150,000 samples of Vietnamese legal question-answering data (vnLawQPA-pos).

The purpose of fine-tuning BartPho with several subsets in P3 is as follows:

- The text summarization dataset helps the model improve its ability to read and effectively interpret text.
- The answer extraction dataset helps the model effectively interpret the input text and identify the correct answer passage within the text.
- The answer selection dataset in P3 helps the model distinguish between correct and incorrect answers.
- The input-based context generation dataset: This dataset trains the model ability to generate text.

Fine-tuning BartPho with the subsets in P3 enables the model to perform basic Vietnamese natural language comprehension tasks. After fine-tuning with P3, we then fine-tuned the model with the vnLawQA-pos dataset to help the model learn how to answer Vietnamese legal questions. Training with P3 first and then vnLawQPA helps the model learn to effectively interpret and generate general text first, and then learn to answer legal questions.

Scenario 3: Fine-tune the Bartpho model with 300,000 samples in the text summarization, answer extraction, and answer selection subsets of P3. Then fine-tune it with 150,000 Vietnamese legal question-answering data vnLawQA-pos.

Upon analyzing the responses generated by the model in Scenario 2 (mentioned in Section 4.2), we observed several outputs were not supported by the input context. We believe this is due to the fine-tuning with the context generation dataset. This dataset aims to generate additional sentences based on 1-2 seed sentences in the input. Consequently, training on this dataset enables the system to generate information that is not present in the input. To validate our assumption, in Scenario 3, we excluded the context generation dataset from BartPho’s fine-tuning process. The experimental results in Section 5 confirm our predictions about the negative impact of the context generation subset. The system’s accuracy improved when this subset was removed from the fine-tuning dataset.

Scenario 4: Fine-tune the Bartpho model with 300,000 samples in the text summarization, answer extraction, and answer selection subsets of P3. Then fine-tune it with 200,000 Vietnamese legal question-answering data vnLawQA.

In practice, there are cases where the legal database or the top-k results returned by the search engine do not contain the answer. In such cases, the input text for the answer generator lacks the necessary information to address the question. In these situations, the system should indicate that no answer exists rather than fabricating a response. The model generated under Scenario 3 did not adequately handle these cases. For Scenario 4, we employed the vnLawQA dataset, which combines 150,000 positive samples with 50,000 negative ones, to train the system to manage such situations correctly.

Scenario 5: Fine-tune the Bartpho model with a generic dataset including 200,000 samples of the vnLawQA, 10,000 samples of the vnGeneralService, 40,000 samples of the DISC-LawLLM, 67,000 samples of BactrianX and 75,000 samples of Vi-Alpaca.

Table 1. Summary of Parameters During Fine-Tuning BartPho and VinaLlama Models

Scenario	Model	Size	Full Finetune	Continual pretraining	Training time (pretrain + finetune)	Testing time
1	BartPho	420M	Yes	Yes	72H + 10H	4H
2	BartPho	420M	Yes	Yes	72H + 30H	4H
3	BartPho	420M	Yes	Yes	72H + 25H	4H
4	BartPho	420M	Yes	Yes	72H + 40H	4H
5	BartPho	420M	Yes	Yes	72H + 35H	4H
6	VinaLlama	7B	QLora	Yes	200H+ 40H	8H
7	VinaLlama	7B	QLora	Yes	200H+120H	8H
8	VinaLlama	7B	QLora	Yes	200H+110H	8H
9	VinaLlama	7B	QLora	Yes	200H+150H	8H
10	VinaLlama	7B	QLora	Yes	200H+140H	8H

As mentioned previously, the vnLawQA dataset, generated by collecting question-answer pairs from the Internet, lacks the diversity to cover various question types. The P3 dataset also have two weakness: (i) it does not contain any answer generation dataset; (ii) the quality of translated version of the P3 dataset is not good. Therefore, in Scenario 5, instead of fine-tuning with the P3 dataset, we propose to use other datasets, including vnGeneralService, DISC-LawLLM, BactrianX, and Vi-Alpaca. In addition, when manually evaluating the results generated by the system in Scenarios 2, 3, 4, we noticed that the larger the finetune dataset in the last step, the more the system forgets the tasks that have been fine-tuned before. This often happened when training low-parameter models like BartPho. So, in Scenario 5, instead of sequentially finetuning each dataset, we mixed all the datasets and finetuned once.

Scenarios 6, 7, 8, 9, and 10 are conducted in a similar manner to scenarios 1, 2, 3, 4, and 5, respectively, with the Bartpho model replaced by the VinaLlama model. Due to the large size of the VinaLlama model, we had to compress its parameter set using Qlora, continual pre-training with the vnLawData dataset, and then fine-tuning with labeled datasets.

4 Experiments

We trained the models following the above-mentioned scenarios on a 3090 GPU with 24GB VRAM. The information related to the model training process is shown in Table 1 as follows:

Since the BartPho pretrained and VinaLlama-7B pretrained models have not been trained for the QA task, they lack the capability to perform QA. VinaLlama-7B-chat [23], a fully fine-tuned version of VinaLlama-7B trained on 1 million samples, including question-answer pairs, is well-suited for handling QA tasks. We conducted experiments using VinaLlama-7B-chat to compare its performance with our models.

Llama3 8B is the most capable openly available LLM to date, which was pre-trained on high-quality datasets in multiple languages, including Vietnamese. It excels particularly in language comprehension, reasoning, and question-answering. Therefore, we compared our models with Llama3 8B which is considered as the gold standard for the Vietnamese legal QA system. In our experiment, the input to Llama3 was a question and relevant text passages from our test set, and the output was an answer.

4.1 Evaluation Measures

To evaluate the system performance, we rely on three measures: Rouge-1, Rouge-2, and Rouge-L [19] to measure the overlap between the system-generated answer and reference answers.

Table 2. Template Used to PROMPT GEMINI for Scoring Answer

Evaluate the following answer to a question on a scale of 0 to 10 based on the following criteria:

- Accuracy: The answer is accurate and complete.
- Conciseness: The answer is concise.
- Clarity: The answer uses clear and easy-to-understand language.
- Coherence: The answer has a logical and coherent structure.
- Grammar correctness: The answer has corrected spelling and grammar.
- Overall score

Text:

Question:

Answer:

Table 3. Experimental Results of all Scenarios Using the Rouge score

Model	Scenario	Rouge-1	Rouge-2	Rouge-L
BartPho	1	33.34%	19.33%	32.21%
BartPho	2	45.12%	33.81%	45.32%
BartPho	3	50.73%	37.23%	51.27%
BartPho	4	56.24%	39.41%	56.11%
BartPho	5	59.82%	48.97%	56.59%
VinaLlama	6	50.12%	35.49%	42.20%
VinaLlama	7	51.32%	38.81%	42.46%
VinaLlama	8	53.54%	39.24%	45.30%
VinaLlama	9	54.91%	39.71%	52.60%
VinaLlama	10	56.52%	39.88%	53.05%
VinaLlama-7B-chat		54.79%	37.31%	52.38%
Llama3		59.23%	48.88%	56.34%

Rouge-N (with N equals 1 or 2) measures the N-gram overlap (where N is the number of words) between the system answer and the reference answers. ROUGE-L measures the **Longest Common Subsequence (LCS)** between the system answer and the reference answers.

However, the Rouge measure cannot handle cases where the system’s response may be phrased differently but still conveys the same meaning as the correct answer. Therefore, we use an additional evaluation by Gemini [6]. Gemini is a large language model developed by Google, capable of generating text in natural language, creating diverse types of content, and responding to user queries in almost any field. Due to its ability to provide good answers based on context and assess the quality of answers based on multiple criteria, we use it to evaluate our two best modules in Scenarios 5 and 10.

Gemini evaluated the answers based on the question and the context. Table 2 shows the template used to prompt Gemini for scoring answers.

4.2 Results and Analysis

Table 3 presents the Rouge scores for our experimental results on 2,000 samples of our test set using ten models generated by the ten scenarios in Section 3.3, the VinaLlama-7B-chat model, and the Llama3 model. It is important to note that our models were fine-tuned on the pre-trained

VinaLlama model [23] after quantization, while the VinaLlama-7B-chat was fully fine-tuned on the original pre-trained VinaLlama model.

The results from Scenario 1 indicate that while the system can generate answers, its accuracy is relatively low. When examining the outputs, we observed that the primary issue arises from a lack of relevant information in the input context, leading to incorrect answers. In these situations, the model tends to hallucinate answers. This issue arises because BARTPho can leverage general knowledge and linguistic patterns to predict subsequent words, leading to the generation of inaccurate or irrelevant answers. To address this, we have developed the vnLawQPA-neg dataset (described in Section 3.2), which includes examples where the context does not contain an answer. In these cases, the system is expected to respond with: "Sorry, I don't have information to answer your question." The effectiveness of this dataset was demonstrated in Scenarios 4, 5, 9, and 10, where the system no longer generated fabricated answers.

Scenario 2 yielded better results than Scenario 1. However, we observed that the model produced some answers that were not grounded in the provided context. Recognizing that the issue was associated with the "input-based context generation" dataset in the P3 dataset, we excluded this dataset from the fine-tuning process in Scenario 3.

Scenario 3 demonstrated further improvement, with Rouge-1, Rouge-2, and Rouge-L scores increasing by 5.61%, 3.42%, and 5.95%, respectively, compared with Scenario 2. The system no longer generated answers that were not based on the provided context. That is consistent with our observation that the "input-based context generation" dataset is unsuitable for training a QA system. Removing it from the fine-tuning process helped the system reduce the inclusion of out-of-context information in the answer.

With the addition of 50,000 negative samples to the training dataset in Scenario 4, we see that the Rouge measures all increased compared with Scenario 3, with Rouge-1, Rouge-2, and Rouge-L rising by 5.51%, 1.98%, and 4.84%, respectively, proving that negative samples play a crucial role in training the system to differentiate between accurate and nearly accurate information, as well as to avoid fabricating answers.

The system performance in Scenario 5 increased significantly with Rouge-1, Rouge-2, and Rouge-L increasing by 3.58%, 9.76%, and 0.48%, respectively, compared with Scenario 4, although the size of augmented datasets in Scenario 4 is larger than that in Scenario 5. It proved that the quality of the training data was more important than its size.

Observing the outputs in Scenario 5, we saw that BartPho often extracted the answer from the context instead of rewriting it to simplify it. This aligns with the characteristic of the vnLawQA dataset, where a high proportion of answers are in extractive form.

In Scenario 1, where the BartPho model was fine-tuned using our domain-specific dataset, the system was still able to generate answers based solely on this dataset. However, its accuracy remained low. In contrast, in Scenarios 2 to 5, when external datasets were incorporated, accuracy improved significantly. This improvement can be attributed to two main reasons. First, the domain-specific dataset was relatively small. Second, it primarily focused on question-answering and lacked the diversity necessary for a deeper comprehension of questions and textual contexts.

The results obtained with the VinaLlama model further highlight the influence of augmented datasets on fine-tuning question-answering systems, consistent with the findings from the experiments using the BartPho model. However, we found that the BartPho model benefited more from augmented training data compared with the VinaLlama model. This can be attributed to the fact that VinaLlama, being larger and more extensively pre-trained, already possesses a strong ability to understand and process text. As a result, while its performance improved with additional data, the gain was not as significant as that of the BartPho model.

Table 4. Experimental Results of the Best BartPho and VinaLlama Models, and Llama3 Using Semantic Scores Given by Gemini

Model	Accuracy	Conciseness	Clarity	Coherence	Grammar correctness	Overall score
BartPho	8.04	7.27	8.30	8.01	8.52	7.72
VinaLlama	7.77	7.56	8.03	7.69	8.22	7.79
Llama3	8.87	9.07	8.81	9.37	8.15	8.75

Table 3 shows that VinaLlama-7B-chat outperforms our VinaLlama models in scenarios 6, 7, and 8. This can be attributed to two main factors: (i) VinaLlama-7B-chat was fully fine-tuned on the original pre-trained VinaLlama model, while our models were fine-tuned on the quantized version, which resulted in reduced accuracy; (ii) the fine-tuning dataset for VinaLlama-7B-chat is 1 million samples, which is larger than ours, covering a broader range of tasks, including question-answering.

In scenarios 9 and 10, our VinaLlama models surpass VinaLlama-7B-chat, indicating that the domain and quality of the training dataset have a substantial impact on system performance.

As shown in Table 3, the Rouge score of the VinaLlama model in Scenarios 10 is lower than that of the BartPho model in Scenarios 5. When looking at the answers generated by the system, we found two reasons:

- VinaLlama often rewrites answers, sometimes using words not from context, but the answer is still correct.
- VinaLlama tends to give more concise answers than BartPho, often cutting ideas out of context.

For those two reasons, when using Rouge measures based on word matching, the accuracy is reduced.

Table 3 shows that our best BartPho model has a slightly higher Rouge score than Llama3, demonstrating that BartPho provides accurate results, comparable to a large and powerful model like Llama3. BartPho achieving higher scores than Llama3 means that BartPho matches more words with the reference answers than Llama3 does, as the BartPho model tends to produce extractive results.

To compare the answers of the best BartPho and VinaLlama models with Llama3 based on semantics, we relied on Gemini’s evaluation results, which were given in the following table:

Observing the output of the BartPho model compared with VinaLlama, we found that VinaLlama generated rewritten answers based on reading comprehension to create concise, easy-to-understand responses that are more aligned with the user’s question, rather than copy text from the input passages as BartPho does in most of the time. Consequently, VinaLlama achieves higher conciseness but lower accuracy compared with BartPho. Additionally, VinaLlama occasionally produces errors such as typos and incomplete sentences due to its pretrained model’s limitations, resulting in poorer performance in terms of clarity, coherence, and grammatical correctness compared with BartPho.

Table 4 shows that Llama3 outperforms both BartPho and VinaLlama in terms of semantic interpretation, highlighting its effectiveness as a model. However, Llama3 requires high-performance GPUs, making it impractical for deployment. Although our model performs slightly worse than the current state-of-the-art model but is more compact and suitable for practical applications.

5 Conclusions and Future Work

In this article, we aim at addressing the problem of developing a Vietnamese legal question-answering system using small-scale LLMs for practical applications. We developed an answer generator in the legal domain based on a Vietnamese legal QA dataset, which we compiled by crawling data from the Internet. To enhance system performance, we enriched the training dataset by translating existing datasets related to text summarization and question answering from English to Vietnamese, and by gathering additional Vietnamese legal QA pairs from the web. We proposed, analyzed, and evaluated various fine-tuning scenarios to identify the optimal approach. Future work will focus on generating a large-scale, high-quality QA dataset to fine-tune our models. Additionally, finding a large-scale reading comprehension dataset that requires reasoning for fine-tuning the QA system is a key aspect of our future efforts to enhance the system's question-answering capabilities.

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A Appendices

A.1 A Data Sample in the vnLawQPA-pos Dataset

The translated version of a data sample in the vnLawQPA-pos dataset:

Question: Can the consulting package valued at 122 million VND, which is part of an investment project for information technology application using recurrent funding, be subject to direct contracting according to Circular 58/2016/TT-BTC?

Passage:

Circular 58/2016/TT-BTC details the use of state funds for procurement to maintain regular operations of state agencies, units of the people's armed forces, public service units, socio-political professional organizations.

Article 3. Forms of Contractor Selection

- (1) Forms of contractor selection include: open bidding, limited bidding, direct contracting, direct procurement, competitive offering, self-implementation, and contractor selection in special cases.
- (2) Based on the annual budget estimate, additional estimates for the year allocated by competent authorities, and the procurement decision of the competent authority as specified in Clause 1, Article 5 of this Circular, the head of the agency or unit shall apply the appropriate form of contractor selection to organize the procurement of goods and services in accordance with regulations.
- (3) For procurement packages of assets, goods, and services that qualify for non-bidding forms of procurement; if the agency or unit deems it necessary to organize bidding to ensure effective management and use of state budget funds, they shall organize open bidding in accordance with regulations and report the results to the competent authority.

Circular 58/2016/TT-BTC details the use of state funds for procurement to maintain regular operations of state agencies, units of the people's armed forces, public service units, political organizations, socio-political organizations, professional socio-political organizations, social organizations, and professional social organizations.

Article 15. Cases of Direct Contracting

- (2) Procurement packages for assets, goods, and services under a project or recurrent procurement estimate with a package value not exceeding 100,000,000 VND (one hundred million VND).
- (3) The implementation of direct contracting for packages specified in Points b, c, d, Clause 1, and Clause 2 of this Article must meet the following conditions:
 - (a) An approved contractor selection plan;
 - (b) The duration of direct contracting from the date of approval of the request dossier to the date of contract signing must not exceed 45 days; for large-scale or complex packages, not more than 90 days;

- (c) The contractor proposed for direct contracting must be listed in the contractor database as stipulated by the Ministry of Planning and Investment (except for packages specified in Clause 2 of this Article).

Decree 63/2014/ND-CP provides detailed regulations on the implementation of some articles of the Bidding Law regarding contractor selection.

Article 54. Direct Contracting Limits

Packages with values within the limits eligible for direct contracting as specified in Point e, Clause 1, Article 22 of the Bidding Law include:

- (1) Not exceeding 500 million VND for consulting services, non-consulting services, public services; not exceeding 1 billion VND for the procurement of goods, construction, mixed packages, purchase of drugs, medical supplies, and industrial products;
- (2) Not exceeding 100 million VND for packages under recurrent procurement estimates.

Answer: According to Clause 2, Article 15 of Circular 58/2016/TT-BTC, procurement packages for assets, goods, and services under a project or recurrent procurement estimate with a package value not exceeding 100,000,000 VND (one hundred million VND) are eligible for direct contracting. Since the information technology application investment package is valued at over 100 million VND, it cannot be subject to direct contracting according to Circular 58/2016/TT-BTC.

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